

Question Paper

Exam Date & Time: 24-Dec-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

SEVENTH SEMESTER B.TECH. DEGREE MAKEUP EXAMINATIONS - DECEMBER 2025

Semantic Web [CSE_4007]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze and examine the relationships among the different layers of the Semantic Web architecture, focusing on how XML, RDF, and OWL interact, and how the Proof and Trust layers function to ensure reliability and confidence. (5)
A)

Course Outcome: "CO1" / Cognitive Level: "Analysis"
- B) Develop and apply a conceptual model showing how the Semantic Web extends the existing Web while maintaining downward compatibility and upward partial understanding between its layers. (5)

Course Outcome: "CO2" / Cognitive Level: "Application"
- 2) Utilize examples from AI or NLP to show how ontologies support reasoning. (4)
A)

Course Outcome: "CO2" / Cognitive Level: "Application"
- B) Classify taxonomies, thesauri, and ontologies based on their structure, relationships, and purpose in knowledge representation. (2)

Course Outcome: "CO2" / Cognitive Level: "Understand"
- C) Analyze the importance of formal semantics in OWL. (4)

Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 3) Apply the concept of building engineering ontology and explain Semantic Sensor Network ontology by giving a diagrammatic overview of the SOSA/SSN ontology modules. (3)
A)

Course Outcome: "CO3" / Cognitive Level: "Application"
- B) Analyze the waterfall methontology in terms of its basic methodological steps, additional tasks and domains where it can be applied. Also draw a diagram for development process and life cycle of methontology. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"
- C) Define re-engineering and describe its process with the help of a diagram. (2)

Course Outcome: "CO4" / Cognitive Level: "Understand"
- 4) Analyse the primary functions and underlying structures of the Systematized Nomenclature of

- A) Medicine-Clinical Terms (SNOMED-CT) and the Unified Medical Language System (UMLS).

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Consider a scenario where a Personal Assistant semantic agent is tasked with curating a list of research articles relevant to a user's specific field of interest, drawing structured metadata from a Semantic Wiki platform (e.g., Semantic MediaWiki). Applying the principles of Semantic Agents and Agent Communication, describe the necessary five steps that agent must execute. (5)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5) Analyze the fundamental structural difference between the Simple Dublin Core standard and the Warwick Framework, and explain how this difference directly impacts their utility in complex digital environments. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- B) Analyze the inherent relationship between the structural requirements of a centralized, open knowledge base like Wikidata and the operational challenges faced by Semantic Web Agents that utilize this data. (5)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4007:Semantic Web

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	15
3	3	CO 3	13
4	4	CO 4	17
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-MIT BLR-MAKEUP EXAM-2025-2026 CSE_4007:Semantic Web

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	4(8%)
3	Application	4	17(34%)
4	Analysis	6	29(58%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		12	50 (100%)